

Work Experience

Chief Technology Officer (CTO), Simunto | Zurich, Switzerland Nov 2024 – Present

- Lead technical strategy, product design, and user experience of Tramola, a web platform for editing, simulating and visualising large scale transport simulations.
- Establish engineering best practices including user-driven design, test-driven development, and systematic testing, fostering a culture of quality and continuous improvement.
- Deliver advanced geospatial analytics capabilities, including big-data heatmaps (Spark, DuckDB), interactive time–distance plots (d3.js), KPI dashboards, and visual analytics components inspired by kepler.gl (deck.gl) and Tableau.

Full Stack Software Developer, Simunto | Zurich, Switzerland Sep 2021 – Nov 2024

- Pioneered adoption of user-driven design and test-driven development, raising product quality and team delivery confidence.
- Optimised Java backend memory footprint, binary data formats, and frontend performance (WebWorkers and reactive processing) to enable efficient visualization of large-scale geospatial data.
- Built a distributed job scheduling framework that abstracts away Slurm/AWS complexity, giving users a one-click “start” button for large-scale transport simulations.

Software Developer Consultant, ThoughtWorks | Berlin, Germany Jan 2020 – Sep 2021

- Operated and maintained 8 Java microservices with a team of 5 on Kubernetes/AWS, handling up to 1,000 requests per second in the payments domain.
- Refactored and improved reliability of a shopping cart component, increasing availability from 99.0% to 99.5% in a highly distributed Scala-based e-commerce platform.
- Contributed to company knowledge sharing by mentoring new starters and running Java practice sessions through pair programming and coding platforms.

Education

ETH Zürich | M. Sc. Computer Science Sep 2021 – Jul 2024
Major in Machine Intelligence, Minor in Data Management Systems

Otto-Friedrich University Bamberg | B. Sc. Applied Computer Science Sep 2016 – Sep 2019
Graduation with distinction (GPA 4.0/4.0 – very good with distinction)

Publications & Projects

Finding Needles in Document Haystacks | CHI 2025

Interactive Explainable for CNNs | IEEE VISxAI 2023

Visual Analytics Framework | IEEE Visualization Workshop 2022

Skills

Programming & Data	Java, Typescript, Javascript, Python, Scala, R, SQL, Spark, Parquet
Web & Visualization	Angular, React, Vue, Redux, D3.js, deck.gl, Maplibre GL, WebWorkers, UX, interactive analytics
Cloud & DevOps	AWS, Kubernetes, CI/CD, Docker, distributed job scheduling, scalable microservices
Practices	Agile, TDD, CI/CD, Pair Programming, Non-violent Communication
Languages	German (native), English (proficient), French (basic)